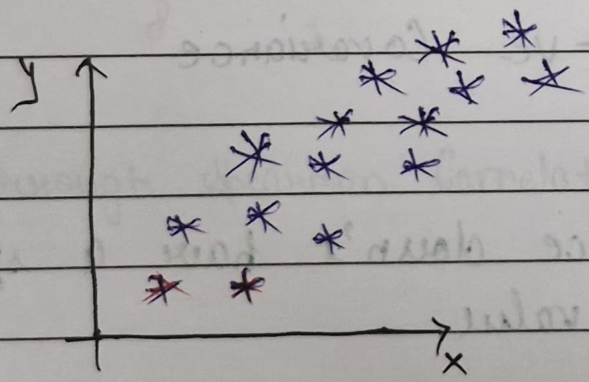
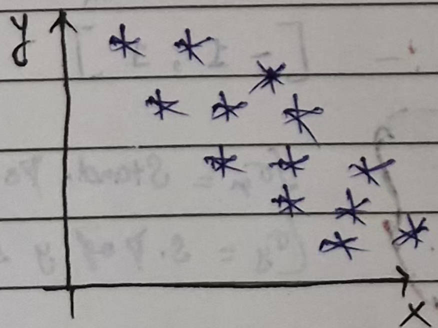
~: COVARIANCE & CORRELATION :~

#→ Relationship b/w x & y



$x\uparrow \quad y\uparrow$

$x\downarrow \quad y\downarrow$



$x\uparrow \quad y\downarrow$

$x\downarrow \quad y\uparrow$

Covariance :-

$$Cov(x,y) :- \sum_{i=1}^{n} \frac{(x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

$$Var(x) = \sum_{i=1}^{n} \frac{(x_i - \bar{x})^2}{n-1}$$

$$= \sum_{i=1}^{n} \frac{(x_i - \bar{x})(x_i - \bar{x})}{n-1}$$

x_i - Data points of x

$\bar{x}$ - Sample mean of x

y_i - Datapoints of y

$\bar{y}$ - Sample mean of y

$Cov(X,X)$

Summary :-

$x \uparrow \; y \uparrow$
$x \downarrow \; y \downarrow$ } +ve Covariance

$x \uparrow \; y \downarrow$
$x \downarrow \; y \uparrow$ } -ve Covariance

Disadvantage :- Co-variance doesn't have a specific limit value

how we will fix

Pearson Correlation Coefficient :- $[-1, 1]$

Min Max

$$\boxed{\rho_{x,y} = \frac{Cov(x,y)}{\sigma_x \cdot \sigma_y}}$$

σ_x = Stand. Deviation of x

σ_y = S. D of y series

★ The more the value towards +1, more +ve Correlated it is (x, y)

★ The more the value towards -1, the more -ve correlated it is (x, y)

Imp:- $Cov(x+y, x-y)$ can be write as :-

$$Cov(x,x) + Cov(y,x) - Cov(x,y) - Cov(y,y)$$

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Spearman Rank Correlation :- [-1, 1]

$$\boxed{\underbrace{r_s}_{\text{rank of } s} = \frac{Cov(R(x), R(y))}{\sigma(R(x)) \cdot \sigma(R(y))}}$$

$\left\{ \begin{array}{l} R(x) \rightarrow \text{Rank of } x \\ R(y) \rightarrow \text{Rank of } y \\ \sigma \rightarrow \text{Standard Dev.} \end{array} \right\}$

Example :-

X	Y	R(X) (highest value in X)	& R(Y)
1	2	5	5
3	4	4	4
5	6	3	3
7	8	2	1
0	7	6	2
8	1	1	6

→ Through Spearman Correlation, we can better find relationship.